



Strategic Analytics of Artificial Intelligence Investment, Financial Globalization, and Environmental Sustainability: Evidence from the United States under the Load Capacity Framework

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Received: 01-16-2026

Revised: 03-15-2026

Accepted: 03-25-2026

Citation: M. O. Farukh, T. Tasnuva, S. I. Tithi, M. A. H. Pabel, and K. Mo, "Strategic analytics of artificial intelligence investment, financial globalization, and environmental sustainability: Evidence from the United States under the load capacity framework," *J. Oper. Strateg. Anal.*, vol. 4, no. 1, pp. 58–68, 2026. <https://doi.org/10.56578/josa040105>.



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Abstract: Environmental sustainability remains a major challenge in the era of digital transformation and global financial integration. The increasing adoption of artificial intelligence (AI) technologies and the expansion of financial globalization continue to reshape economic systems and influence ecological outcomes. This study investigates the dynamic relationships among private AI investment, financial globalization, and environmental sustainability in the United States within the Load Capacity Curve (LCC) framework and from a strategic analytics perspective. Annual time-series data from 1990–2019 were employed. Economic growth, technological innovation, and urbanization were incorporated as additional determinants of environmental sustainability measured by the load capacity factor (LCF). Unit root procedures were conducted, and the Autoregressive Distributed Lag (ARDL) bounds testing framework was applied to estimate both long-run equilibrium relationships and short-run dynamics. Robustness analysis was further performed using alternative cointegration estimators. The results showed that a long-run equilibrium relationship existed among the variables. A U-shaped relationship between income and environmental sustainability was identified, supporting the LCC hypothesis. Private investment in AI positively affected ecological capacity, suggesting that AI-related investment contributed to environmental improvement through resource optimization and efficiency gains. Financial globalization and technological innovation negatively affected environmental sustainability, implying that uncontrolled financial expansion and non-green technological activities intensified ecological pressure. Urbanization demonstrated a positive long-run contribution to ecological sustainability. The robustness analysis produced consistent findings. The results indicate that AI-related investment can serve as a strategic instrument for balancing technological development and ecological objectives. This study provides evidence that integrating strategic analytics with sustainability assessment improves understanding of the environmental implications of digital transformation. The findings offer practical decision support for policymakers seeking to align technological investment and global financial integration with long-term sustainability goals.

Keywords: Strategic analytics; Artificial intelligence investment; Financial globalization; Load capacity factor; Environmental sustainability; Decision support systems; United States

1 Introduction

Environmental degradation and climate-related risks have emerged as central challenges to sustainable development in the twenty-first century [1]. Rapid industrialization, excessive resource utilization, and growing energy demand have intensified pressure on ecological systems, leading to biodiversity loss, declining environmental quality, and increased vulnerability to climate change [2, 3]. Traditionally, environmental performance has been assessed using indicators such as carbon dioxide emissions and ecological footprint [4, 5]. However, these measures often provide a partial view of ecological sustainability, as they primarily capture environmental pressure without adequately reflecting the regenerative capacity of natural systems. In response, recent literature has increasingly emphasized the

relevance of the load capacity factor (LCF), which integrates both ecological footprint and biocapacity to offer a more comprehensive assessment of environmental sustainability [6, 7]. A higher LCF indicates that a nation's ecological resources are sufficient to support its consumption patterns, whereas a lower value signals ecological imbalance and unsustainable resource use [8]. Therefore, examining environmental dynamics through the lens of LCF provides deeper insights into the sustainability of economic and technological transitions, particularly in advanced economies where environmental pressures are closely linked with structural transformation and innovation-driven growth.

The United States represents a critical case for examining the nexus between economic progress and environmental sustainability due to its significant contribution to global output and environmental pressure. As one of the largest economies in the world, the country has experienced sustained economic expansion alongside increasing technological advancement and urban development [9, 10]. However, this growth trajectory has also been accompanied by substantial environmental challenges, including high energy consumption and considerable greenhouse gas emissions. Despite improvements in environmental regulations and the adoption of cleaner technologies, the ecological balance remains under pressure due to rising consumption patterns and industrial activities [11, 12]. In this context, understanding the determinants of ecological sustainability becomes essential for designing effective policy interventions. Factors such as financial globalization, technological innovation, and urbanization have reshaped the economic structure of the United States, influencing both production processes and environmental outcomes [13, 14]. At the same time, the growing integration of artificial intelligence (AI) into economic systems introduces new dimensions to the sustainability debate, as it has the potential to enhance resource efficiency while also increasing energy demand [15]. Therefore, a comprehensive assessment of these interconnected dynamics is crucial to evaluate whether the United States can achieve long-term environmental sustainability alongside continued economic growth.

The evolving role of technological transformation and global economic integration has intensified scholarly interest in understanding how modern drivers influence environmental sustainability. Among these, private investment in AI has emerged as a pivotal factor shaping production efficiency, energy management, and environmental monitoring systems [16, 17]. AI-driven technologies can optimize resource allocation, reduce waste, and improve environmental governance through data-driven decision-making [18]. However, the rapid expansion of AI infrastructure and computational requirements may also increase energy consumption, potentially offsetting its environmental benefits [19]. Similarly, financial globalization facilitates cross-border capital flows, enabling investments in advanced technologies and green infrastructure, yet it may also encourage pollution-intensive industrial relocation and resource exploitation [20, 21]. Technological innovation, while essential for long-term productivity, presents a dual effect by either enhancing environmental efficiency or accelerating ecological degradation depending on its orientation [22]. Urbanization further complicates this relationship, as expanding urban centers increase demand for energy, transportation, and infrastructure, thereby influencing environmental outcomes [23]. These interrelated dynamics highlight the necessity of adopting an integrated strategic analytics framework capable of supporting data-driven policy evaluation and long-term decision-making concerning economic growth, digital transformation, and environmental sustainability, particularly when assessed through a comprehensive indicator such as the LCF.

To systematically examine these relationships, this study is grounded in the Load Capacity Curve (LCC) hypothesis, which extends the traditional environmental Kuznets curve by incorporating a more comprehensive measure of ecological sustainability. The LCC framework postulates a non-linear relationship between economic growth and the LCF, typically characterized by a U-shaped pattern. In the early stages of economic development, increased production and resource utilization tend to exert significant pressure on the environment, thereby reducing ecological capacity [24, 25]. However, as income levels rise, economies are more likely to adopt cleaner technologies, enforce environmental regulations, and shift toward sustainable production practices, leading to improvements in ecological balance [26]. This theoretical perspective provides a robust foundation for analyzing how modern economic drivers interact with environmental systems. Within this context, variables such as AI investment, financial globalization, technological innovation, and urbanization are incorporated to capture the multifaceted nature of contemporary economic transformation. By embedding these factors within the LCC framework, the study offers a more nuanced understanding of how technological and financial forces shape environmental sustainability in an advanced economy like the United States, where digitalization and globalization play increasingly dominant roles.

Despite the growing body of literature on environmental sustainability, several critical gaps remain unaddressed. First, existing studies predominantly rely on conventional indicators such as carbon emissions or ecological footprint, which fail to fully capture the balance between ecological demand and regenerative capacity [27, 28]. The application of the LCF remains relatively limited, particularly in the context of advanced economies like the United States. Second, while the environmental implications of economic growth, globalization, and technological innovation have been widely explored, the role of private investment in AI has received comparatively little empirical attention. Given the rapid expansion of AI-driven systems and their potential to reshape production and environmental management processes, this omission represents a significant gap in the literature [29]. Third, prior studies often examined these determinants independently and paid limited attention to their implications from a strategic analytics and decision-support perspective. Existing studies rarely provided an integrated framework capable of supporting

data-driven sustainability decisions under conditions of technological and economic transformation.

Furthermore, limited evidence exists regarding how analytical frameworks can support strategic environmental decision-making in digitally transformed economies. Understanding these relationships is increasingly important for policymakers and institutions seeking evidence-based approaches to balance technological advancement with sustainability objectives.

To address these limitations, this study contributes to the literature in several ways. It adopts the LCF as a comprehensive sustainability indicator, incorporates private AI investment as a key explanatory variable, and examines the combined effects of financial globalization, technological innovation, and urbanization within a unified framework. By focusing on the United States, the study provides context-specific insights and extends existing literature by introducing a strategic analytics perspective for environmental decision support under digital transformation.

2 Literature Review

The measurement of environmental sustainability has evolved significantly over time, reflecting the growing complexity of ecological challenges associated with economic development. Traditional indicators such as carbon dioxide emissions and ecological footprint have been widely used to assess environmental degradation; however, these measures often provide a unidimensional perspective by focusing primarily on environmental pressure. In contrast, the LCF has emerged as a more comprehensive indicator, as it captures the balance between ecological footprint and biocapacity. By integrating both the demand placed on natural resources and the ecosystem's regenerative ability, LCF offers a more holistic understanding of environmental sustainability. A higher LCF indicates that a country possesses sufficient ecological capacity to sustain its consumption patterns, whereas a lower value reflects ecological deficit and unsustainable resource use [7, 30]. This dual perspective allows researchers to evaluate not only the extent of environmental degradation but also the resilience of natural systems. Consequently, recent studies have increasingly adopted LCF as a preferred measure to assess ecological sustainability, particularly in the context of dynamic economic transformations. Its ability to reflect both environmental stress and recovery potential makes it especially suitable for analyzing the sustainability implications of modern growth drivers, including technological advancement and global economic integration.

The relationship between economic growth and environmental sustainability has been extensively examined in the literature [31, 32], with increasing attention given to the LCC hypothesis. Unlike the traditional Environmental Kuznets Curve, which primarily focuses on emissions, the LCC framework evaluates how economic expansion influences the balance between ecological demand and regenerative capacity [33, 34]. In the early stages of growth, rising income levels are generally associated with intensified industrial activity, higher energy consumption, and increased exploitation of natural resources, leading to a decline in the LCF [35]. However, as economies mature, structural transformation, technological advancement, and stronger environmental awareness may contribute to improved ecological outcomes. This results in a non-linear, often U-shaped relationship between income and LCF. Empirical findings in this area remain mixed. Several studies report that economic growth initially deteriorates environmental quality by reducing ecological capacity, while others provide evidence supporting the long-term recovery of environmental sustainability as income rises [36, 37]. These divergent outcomes suggest that the impact of economic growth on LCF is highly context-dependent and influenced by factors such as energy structure, policy frameworks, and technological progress. Therefore, further investigation is necessary to clarify the nature of this relationship, particularly within advanced economies.

The rapid expansion of AI has introduced a new dimension to the sustainability discourse, attracting growing academic attention in recent years. AI-driven technologies have the potential to enhance environmental sustainability by improving energy efficiency, optimizing production processes, and enabling real-time environmental monitoring [38, 39]. Through advanced data analytics and automation, AI can facilitate smart resource management, reduce waste generation, and support the transition toward cleaner production systems [40]. For instance, applications such as intelligent energy grids, precision agriculture, and climate modeling demonstrate how AI can contribute to ecological balance. However, the environmental implications of AI are not unambiguously positive [41]. The development and deployment of AI systems require substantial computational power, leading to increased electricity consumption and potential carbon emissions, particularly when energy sources are not renewable [42]. This dual effect creates a paradox where AI simultaneously supports and challenges environmental sustainability. Despite its growing importance, empirical research examining the direct relationship between private investment in AI and ecological indicators such as the LCF remains limited [43]. Most existing studies focus on emissions or energy consumption, leaving a significant gap in understanding how AI investment influences the broader ecological capacity of an economy [44]. This gap highlights the need for more comprehensive and integrated analyses in this area. Moreover, understanding AI investment from a strategic analytics perspective may provide stronger decision-support mechanisms for balancing technological advancement and sustainability objectives.

Financial globalization has become a defining feature of modern economies, shaping investment patterns, technological diffusion, and environmental outcomes. By facilitating cross-border capital flows, financial globalization

can enhance access to advanced technologies and support investments in environmentally sustainable infrastructure [33, 45]. In this context, it may contribute positively to ecological sustainability by enabling cleaner production processes and promoting green finance initiatives. However, the environmental consequences of financial globalization are far from uniform. In some cases, increased financial integration encourages the relocation of pollution-intensive industries to regions with weaker environmental regulations, thereby exacerbating ecological degradation [46]. This phenomenon is often associated with the pollution haven hypothesis, where economic benefits are achieved at the expense of environmental quality. Empirical evidence on the relationship between financial globalization and sustainability remains inconclusive [47]. While some studies report that financial globalization improves environmental performance through efficiency gains and technological transfer, others find that it intensifies environmental pressure due to increased industrial activity and resource exploitation [48, 49]. These conflicting findings suggest that the impact of financial globalization largely depends on the regulatory framework, institutional quality, and the nature of investment flows within an economy. Therefore, a deeper investigation is required to understand its role in shaping the LCF, particularly in highly integrated economies such as the United States. Such investigation may also support evidence-based strategic decisions concerning sustainable financial integration.

Technological innovation plays a central role in shaping the trajectory of environmental sustainability, yet its impact remains inherently ambiguous. On one hand, innovation contributes to the development of cleaner production techniques, energy-efficient systems, and environmentally friendly technologies that can reduce ecological pressure [50]. Advances in renewable energy, waste management, and industrial efficiency have been widely recognized as key drivers of sustainable development [46]. On the other hand, technological progress may also intensify environmental degradation if it leads to increased production, higher energy consumption, and resource overexploitation [51]. This dual effect is often explained through the scale, technique, and rebound mechanisms, where efficiency gains may be offset by expanded economic activity. Empirical studies provide mixed evidence regarding the environmental implications of technological innovation [52]. Some findings suggest that innovation improves environmental quality by reducing emissions and enhancing resource efficiency, while others indicate that it may worsen ecological conditions, particularly when technological advancement is not aligned with green objectives [41]. Moreover, existing research has primarily focused on conventional indicators such as carbon emissions, with limited attention given to broader measures like the LCF [53]. This highlights the need for further investigation into how technological innovation influences ecological sustainability within a more comprehensive analytical and decision-support framework.

Urbanization is widely recognized as a key driver of structural transformation, significantly influencing environmental sustainability through changes in consumption patterns, infrastructure development, and energy demand [4]. The migration of populations from rural to urban areas typically leads to increased industrial activity, transportation needs, and pressure on natural resources, which can adversely affect ecological balance. Expanding urban centers often require extensive infrastructure and energy-intensive services, contributing to environmental degradation and a decline in ecological capacity [18, 49]. However, the relationship between urbanization and environmental sustainability is not uniformly negative. With appropriate planning and technological integration, urbanization can promote more efficient resource utilization through economies of scale, improved public transportation systems, and the development of smart cities [13, 23]. These advancements can reduce per capita energy consumption and environmental impact over time. Empirical evidence on the urbanization–environment nexus remains inconclusive. While several studies report that urban expansion exacerbates environmental degradation, others find that urbanization can enhance sustainability under well-regulated conditions [29]. In the context of the LCF, the effect of urbanization depends largely on the quality of urban planning, technological adoption, and policy frameworks. Therefore, examining this relationship within a comprehensive model is essential to understand its overall impact on ecological sustainability.

Despite the expanding literature on environmental sustainability, several important gaps persist that warrant further investigation. First, most empirical studies rely on conventional environmental indicators such as carbon emissions and ecological footprint, which do not fully capture the balance between ecological demand and regenerative capacity. The use of the LCF remains relatively limited, particularly in studies focusing on advanced economies [7]. Second, while considerable attention has been given to economic growth, globalization, and technological innovation, the environmental implications of private investment in AI remain underexplored. Given the rapid expansion of AI technologies and their transformative impact on production systems, ignoring this dimension may lead to an incomplete understanding of modern sustainability dynamics [45]. Third, existing studies often analyze these determinants independently and provide limited evidence from a strategic analytics perspective. Integrated analytical frameworks capable of supporting sustainability-related decision processes remain relatively scarce. Additionally, empirical evidence specific to the United States remains scarce, despite its significant role in global economic and environmental systems. To address these limitations, this study adopts a comprehensive approach by incorporating AI investment, financial globalization, technological innovation, and urbanization within the load capacity framework. This integrated perspective provides a more nuanced understanding of the drivers of ecological sustainability and

contributes to the ongoing debate on sustainable development.

3 Methodology

This study utilizes annual time-series data for the United States covering the period 1990–2019 to examine the determinants of environmental sustainability. The LCF is employed as the dependent variable to capture ecological balance. The key explanatory variables include private investment in AI, financial globalization, technological innovation, economic growth, and urbanization. All variables are transformed into logarithmic form to ensure consistency and reduce heteroscedasticity. Data are sourced from reliable international databases, including the Global Footprint Network for LCF, the World Development Indicators for macroeconomic variables, Our World in Data for AI investment and innovation, and the KOF Globalization Index for financial globalization. The selected variables were structured to provide a data-driven analytical perspective for evaluating the sustainability implications of technological and economic transformation.

This study adopts a comprehensive econometric framework to examine the dynamic relationship between environmental sustainability and its determinants in the United States. The analysis begins with unit root testing to determine the stationarity properties of the variables. Specifically, Augmented Dickey–Fuller (ADF), Phillips–Perron (PP), and Dickey–Fuller Generalized Least Squares (DF-GLS) tests are employed to ensure that none of the variables are integrated beyond the first order. Establishing the order of integration is essential for selecting an appropriate estimation technique. Given the possibility of mixed integration among variables, the Autoregressive Distributed Lag (ARDL) bounds testing approach is utilized to investigate long-run relationships. The ARDL method is particularly suitable for small sample sizes and allows for the estimation of both short-run dynamics and long-run equilibrium simultaneously. The bounds test is applied to confirm the existence of cointegration among the variables. Once cointegration is established, the long-run coefficients and short-run error correction model (ECM) are estimated, where the error correction term captures the speed of adjustment toward equilibrium following short-term shocks. To validate the robustness of the ARDL results, additional long-run estimators, including Fully Modified Ordinary Least Squares (FMOLS), Dynamic Ordinary Least Squares (DOLS), and Canonical Cointegrating Regression (CCR), are employed. These techniques address potential issues related to endogeneity and serial correlation. Furthermore, the Pairwise Granger causality test is conducted to explore the direction of causal relationships among variables. Finally, a set of diagnostic tests, including the Jarque–Bera test for normality, Breusch–Godfrey LM test for serial correlation, and Breusch–Pagan–Godfrey test for heteroscedasticity, are performed to ensure the reliability and stability of the estimated model. In addition to statistical estimation, the analytical framework supports strategic interpretation of long-term sustainability dynamics and provides a quantitative basis for evidence-based environmental decision processes.

4 Results and Discussion

The descriptive statistics presented in Table 1 provide an overview of the distributional properties of the variables used in the analysis. The results indicate that most variables exhibit moderate variability, as reflected by relatively low standard deviations, suggesting stable trends over the study period. The mean values highlight that economic growth and technological indicators maintain consistently positive levels, whereas the LCF remains comparatively lower, indicating potential ecological pressure. The range between minimum and maximum values shows no extreme outliers, confirming data consistency. Additionally, the distributional properties, including skewness and kurtosis, suggest that the variables generally follow an acceptable pattern for time-series analysis, supporting the validity of subsequent econometric estimations. The initial statistical profile also provides a preliminary basis for subsequent data-driven analytical interpretation.

To ensure the robustness of the empirical analysis, the stationarity properties of the variables are examined using ADF, PP, and DF-GLS tests, as reported in Table 2. The results indicate that most variables are non-stationary at their level form but become stationary after first differencing, confirming integration at order I (1). However, urbanization appears stationary at level, indicating an I (0) process. This mixed order of integration justifies the application of the ARDL bounds testing approach. Overall, the findings in Table 2 confirm that none of the variables are integrated beyond the first order, ensuring the validity of subsequent cointegration analysis.

To examine the existence of a long-run equilibrium relationship among the variables, the ARDL bounds testing approach is applied, and the results are shown in Table 3. The estimated F-statistic exceeds the upper critical bound at conventional significance levels, leading to the rejection of the null hypothesis of no cointegration. This result confirms that a stable long-run relationship exists between the LCF and its determinants, including economic growth, AI investment, financial globalization, technological innovation, and urbanization. The presence of cointegration justifies proceeding with the estimation of both long-run coefficients and short-run dynamics within the ARDL framework, ensuring a comprehensive understanding of the underlying relationships and supporting evidence-based interpretation of sustainability dynamics.

Table 1. Descriptive statistics of variables

Statistic	LCF	LGDP	LGDP2	LPAI	LFGOB	LTI	LURBA
Mean	-0.812	10.721	115.238	21.874	4.315	12.214	4.389
Median	-0.801	10.755	116.502	21.430	4.330	12.298	4.392
Maximum	-0.602	11.210	126.104	25.982	4.402	12.640	4.430
Minimum	-0.965	10.120	102.875	20.310	4.080	11.410	4.325
Std. Dev.	0.088	0.335	7.112	1.702	0.095	0.415	0.029
Skewness	0.102	-0.210	-0.180	0.915	-1.090	-0.540	-0.480
Kurtosis	2.01	1.92	1.89	2.52	3.05	1.95	2.30
Jarque-Bera	1.38	1.85	1.79	5.30	6.85	3.20	2.05
Probability	0.502	0.395	0.410	0.070	0.032	0.201	0.360

Note: LCF = load capacity factor; LGDP = logarithm of gross domestic product (GDP); LGDP2 = square of logarithm of GDP; LPAI = log of primary agricultural investment; LFGOB = log of fiscal government outlay on agriculture; LTI = log of total investment; LURBA = log of urbanization rate/urban population share; Std. Dev. = standard deviation.

Table 2. Unit root test results

Variable	ADF I (0)	ADF I (1)	PP I (0)	PP I (1)	DF-GLS I (0)	DF-GLS I (1)
LCF	-0.912	-5.284***	-0.845	-5.301***	-1.502	-4.215***
LGDP	-0.965	-4.702***	-1.002	-4.685***	-1.658	-3.402***
LGDP2	-0.721	-4.935***	-0.689	-4.910***	-1.774	-3.365***
LPAI	-0.845	-7.120***	-1.765	-7.045***	-0.995	-5.210***
LFGOB	-2.010	-4.002***	-2.045	-3.980***	-1.885	-4.401***
LTI	-1.965	-4.985***	-2.020	-5.010***	-0.932	-3.712***
LURBA	-8.520***	-1.650	-5.002***	-1.702	-3.640***	-1.398

Note: ADF = Augmented Dickey–Fuller; PP = Phillips–Perron; DF-GLS = Dickey–Fuller Generalized Least Squares; LCF = load capacity factor; LGDP = logarithm of gross domestic product (GDP); LGDP2 = square of logarithm of GDP; LPAI = log of primary agricultural investment; LFGOB = log of fiscal government outlay on agriculture; LTI = log of total investment; LURBA = log of urbanization rate/urban population share.

Table 3. Autoregressive Distributed Lag (ARDL) bounds test results

Test Statistic	Value	K
F-statistic	6.128	6
Significance level	I (0)	I (1)
10%	2.05	3.10
5%	2.39	3.38
2.5%	2.70	3.73
1%	3.06	4.15

Following the confirmation of cointegration, the ARDL model is estimated to capture both long-run equilibrium relationships and short-run dynamics among the variables. The results reveal that economic growth exerts a negative and statistically significant impact on the LCF in the long run, indicating that initial stages of economic expansion intensify environmental pressure. However, the positive and significant coefficient of the squared income term confirms the existence of a U-shaped relationship, supporting the LCC hypothesis. This implies that environmental sustainability improves after a certain income threshold, reflecting structural transformation and increased environmental awareness in advanced stages of development. Private investment in AI demonstrates a positive and significant effect on the LCF in both the short and long run. This suggests that AI-driven technologies enhance environmental sustainability by improving resource efficiency, optimizing energy use, and enabling more effective environmental monitoring systems. In contrast, financial globalization exhibits a negative association with ecological capacity, implying that increased financial integration may facilitate environmentally harmful activities, such as pollution-intensive production and excessive resource utilization.

As further shown in Table 4, technological innovation also has a negative impact on the LCF, indicating that current innovation patterns may not be sufficiently aligned with green objectives. This finding highlights the presence of a rebound effect, where efficiency gains are offset by increased production and consumption. Urbanization, on the other hand, displays a positive contribution to environmental sustainability in the long run, suggesting that urban development, when supported by efficient infrastructure and planning, can improve ecological outcomes. In the short

run, the estimated coefficients largely mirror the long-run relationships, although the magnitude of effects differs. The error correction term is negative and statistically significant, confirming the stability of the model and indicating a relatively fast adjustment toward long-run equilibrium following short-term shocks. Overall, the results emphasize the complex interplay between economic growth, digital transformation, and environmental sustainability in the United States and provide analytical insights relevant to long-term sustainability-oriented decision processes.

Table 4. Autoregressive Distributed Lag (ARDL) long-run and short-run results

Variables	Long-Run Coefficient (t-stat)	Short-Run Coefficient (t-stat)
LGDP	-3.215*** (10.842)	-3.978** (4.115)
LGDP2	1.102*** (0.482)	0.148** (0.176)
LPAI	0.021** (0.029)	0.438*** (0.006)
LFGOB	-1.085*** (0.268)	-0.472*** (0.142)
LTI	-0.119** (0.152)	-0.0031 (0.006)
LURBA	1.256* (4.982)	15.874*** (2.105)
ECT (-1)	—	-0.712*** (0.081)
Constant	9.845*** (28.114)	—
R-squared	0.869	—

Note: LCF = load capacity factor; LGDP = logarithm of gross domestic product (GDP); LGDP2 = square of logarithm of GDP; LPAI = log of primary agricultural investment; LFGOB = log of fiscal government outlay on agriculture; LTI = log of total investment; LURBA = log of urbanization rate/urban population share; ECT = error correction term.

5 Conclusion and Policy Recommendations

This study investigated the dynamic relationship between environmental sustainability and its key determinants in the United States within the framework of the LCC hypothesis. Using time-series data and the ARDL approach, the findings provided evidence of a stable long-run equilibrium relationship among the variables. The results confirmed a U-shaped association between economic growth and the LCF, supporting the LCC hypothesis and suggesting that environmental conditions improve after a certain income threshold is reached. Private investment in AI was found to contribute positively to ecological sustainability, indicating its potential role in improving resource efficiency and environmental management practices. In contrast, financial globalization and technological innovation exerted adverse effects on ecological capacity, reflecting the environmental risks associated with uncontrolled financial expansion and non-green innovation activities. Urbanization showed a positive long-run contribution, suggesting that effective planning and infrastructure development can support environmental improvement.

From a strategic analytics perspective, the findings suggest that sustainability outcomes are increasingly shaped by interactions among digital transformation, financial systems, and technological change. The empirical evidence highlights the importance of analytical frameworks capable of supporting evidence-based interpretation of long-term sustainability dynamics. Integrating environmental indicators with analytical decision processes may provide a stronger basis for balancing technological advancement and ecological objectives.

The findings also provide several practical policy implications for environmental management in the United States. First, economic growth alone may not guarantee ecological sustainability unless accompanied by environmentally oriented development strategies. Policymakers may therefore prioritize green growth initiatives through cleaner production systems and carbon mitigation measures. Second, the positive contribution of AI investment suggests that AI-supported technologies can be incorporated into sustainability planning and environmental monitoring systems. Public incentives, targeted investment programs, and collaborative initiatives may strengthen the development of environmentally responsible AI applications. Third, stronger institutional mechanisms may be required to ensure that financial globalization promotes sustainable rather than pollution-intensive activities. Likewise, technological innovation policies should encourage environmentally oriented innovation pathways. Finally, sustainable urban planning and smart infrastructure development may support long-term ecological improvement.

From a managerial and decision-support perspective, the findings offer analytical guidance for policymakers, investment institutions, and organizations involved in sustainability planning. The results indicate that AI investment can serve not only as a technological driver but also as a strategic instrument for supporting long-term environmental decision processes under conditions of digital transformation.

This study has several limitations that provide directions for future research. First, the analysis focused on a single-country setting, which may limit broader generalization. Future studies may expand the framework to multi-country or regional analyses. Second, the proxy used for AI investment may not fully capture the multidimensional nature of digital transformation. Third, interaction effects among variables were not explicitly examined and may provide additional insights into the complex mechanisms influencing environmental sustainability.

Author Contributions

Conceptualization, M.O.F.; methodology, M.O.F.; software, M.O.F.; validation, M.O.F., M.A.H.P., and K.M.; formal analysis, T.T., S.I.T., M.A.H.P., and K.M.; investigation, T.T. and S.I.T.; data curation, M.O.F.; writing—original draft preparation, M.O.F.; writing—review and editing, T.T., S.I.T., M.A.H.P., and K.M.; visualization, M.O.F. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data is available on request from the corresponding author.

Conflicts of Interest

The authors declare no conflicts of interest.

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